

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				Or For one oscillation: $m = 1.97 \pm 10\%$ (1) $\left[T = 2\pi\sqrt{\frac{l}{g}} \right], \therefore \sqrt{g} = \frac{2\pi}{m}$ [or by impl.] [= $3.19 \pm 10\%$] (1) $\therefore g = 10.2 (\pm 20\%) \text{ m s}^{-2}$ Alternative: calculate max value of g from min gradient; min value of g from max gradient and hence $g \pm$ uncertainty. Gradients of both: 43.284 and 35.357 (1) Mean $T_{20} = \frac{40\pi}{\sqrt{g}} \sqrt{l}$ so gradient = $\frac{40\pi}{\sqrt{g}}$ $g = \left(\frac{40\pi}{43.3}\right)^2 = 8.42 \text{ m s}^{-2}$ and $g = \left(\frac{40\pi}{35.3}\right)^2 = 12.67 \text{ m s}^{-2}$ (1) So $\text{unc}(g) = \frac{12.67-8.42}{2} = 2.1 \text{ m s}^{-2}$ (1) $g = 10.5 \text{ m s}^{-2}$ [1] $\therefore \% \text{ unc}(g) = \left[\frac{2.1}{10.5} \times 100\% \right] = 20\%$ (1)						